PROJECT REPORT

A Hybrid, Feature-Optimized Decision-Support Framework for Paddy Yield Prediction and Variety Recommendation

PaddyWise — Yield & Variety Advisor

Submitted by — Group 2

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Abstract

Rice is the staple crop of Tamil Nadu, and farmers must decide which paddy variety to grow and how to manage it before the season begins. This project develops a **hybrid, feature-optimized, decision-support machine-learning framework** on the UCI Paddy Dataset (2,789 farm records, 45 columns, six agri-blocks of Cuddalore/Kallakurichi districts). The framework (1) identifies the key factors influencing paddy yield, (2) predicts rice yield, and (3) recommends the most suitable variety according to field conditions and the farmer's own priority.

After cleaning (451 duplicate rows removed) and exploratory analysis, a **three-stage feature optimization** removed a data-leakage column, ten dosage columns that are exact multiples of farm area, and 28 weather columns that are fully determined by the agri-block, reducing **44 inputs to 5 features** with no loss of accuracy. Five regressors were compared on a hidden 20 % test set and with 5-fold cross-validation. The proposed **Hybrid Stacking model** (two Gradient Boosting models and a Random Forest combined by a non-negative linear meta-model) achieved the best results: **R^2 = 0.9903, MAE = 658 kg, RMSE = 913 kg, MAPE = 2.89 %** (≈ 97.1 % accuracy) and a cross-validated R^2 of 0.9916 ± 0.0011. The hybrid model was also re-implemented from scratch in NumPy and matched scikit-learn to within 0.08 % on average.

Farm area is the dominant factor for total yield; for productivity per hectare the agri-block (location and microclimate), variety, nursery type and soil follow. The decision-support layer predicts the yield of every variety for the farmer's conditions, ranks them with a transparent weighted multi-criteria score, and generates a day-wise input plan (seed, fertilizer, weedicide, pesticide, micronutrients) scaled to the farm. Everything is delivered through an interactive Streamlit web application.

Keywords: paddy yield prediction, stacking ensemble, gradient boosting, random forest, feature selection, decision support system, multi-criteria recommendation.

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1. Introduction

1.1 Background

Paddy (rice) cultivation supports millions of farmers in India. Yield depends on many interacting factors: farm size, location and its weather, the variety grown, soil, nursery method and input management. Farmers usually make these decisions from experience. A data-driven tool that estimates the expected yield and suggests the best variety for a given field can reduce risk and improve income.

1.2 Problem statement

> Develop a hybrid, feature-optimized and decision-support machine learning framework that identifies the key factors influencing paddy growth, predicts rice yield, and subsequently recommends the most suitable paddy variety based on agricultural conditions and farmers' preferences.

1.3 Objectives

1. **Clean and understand** the UCI Paddy Dataset (preprocessing and EDA).
2. **Optimize the feature set**: remove leakage, redundant and dependent columns while keeping accuracy.
3. **Build and compare** several regression models and propose a **hybrid stacking ensemble**.
4. **Test** every model on unseen data with standard metrics and cross-validation.
5. **Identify the dominant factors** that drive yield.
6. **Recommend a variety** using the model's predictions and the farmer's priorities, and generate a **day-wise input plan**.
7. **Deploy** the framework as an easy-to-use web application.

1.4 Meaning of the three key words

Term	What it means in this project
Hybrid	Several different ML models are combined by stacking; a meta-model learns how much to trust each.
Feature-optimized	Only the informative, non-redundant, leakage-free inputs are kept (44 → 5).
Decision-support	The output is not only a number: the system ranks varieties for the farmer's priority and produces an input plan.

2. Dataset description

The dataset was published by M. Subramaniyan on the UCI Machine Learning Repository (2023). Each row is one paddy farm for one season. It contains **2,789 rows and 45 columns**: 44 inputs and the target **Paddy yield (kg)**.

Group	Columns	Examples
Farm & location	4	Hectares (1–6), Agriblock (6 blocks), Variety (3), Soil type (clay / alluvial)
Nursery & land preparation	5	Nursery (wet / dry), nursery area, seed rate, land-preparation inputs
Fertilizer & crop protection	6	DAP @ 20 days, thiobencarb @ 28 days, urea @ 40 days, potash @ 50 days, pesticide @ 60 days, micronutrients @ 70 days
Rainfall (4 periods)	8	rainfall and "AI" values for days 1–30, 31–50, 51–70, 71–105
Temperature (4 periods)	8	min / max temperature for days 1–30, 31–60, 61–90, 91–120
Wind (4 periods)	8	wind speed (knots) and wind direction
Humidity (4 periods)	4	relative humidity
Post-harvest	1	Trash (straw bundles)
Target	1	**Paddy yield (kg)**: 5,410 to 38,814 kg

Variety	Duration (days)	Grain type	Indicative price (₹/kg)	Avg yield in data (kg/ha)
CO 43	135	Medium slender	22	5990
Ponmani	155	Medium bold	21	5962
Delux Ponni	140	Fine	27	6006

Duration, grain type and prices are general published characteristics of these varieties (prices are indicative and editable in the app). They are **not** part of the dataset and are used only in the recommendation step.

3. Methodology: the overall framework

Figure 1 shows the complete pipeline. Each block is explained in its own chapter. The two dark blocks are the core of the project: the **hybrid model** and the **decision-support layer**.

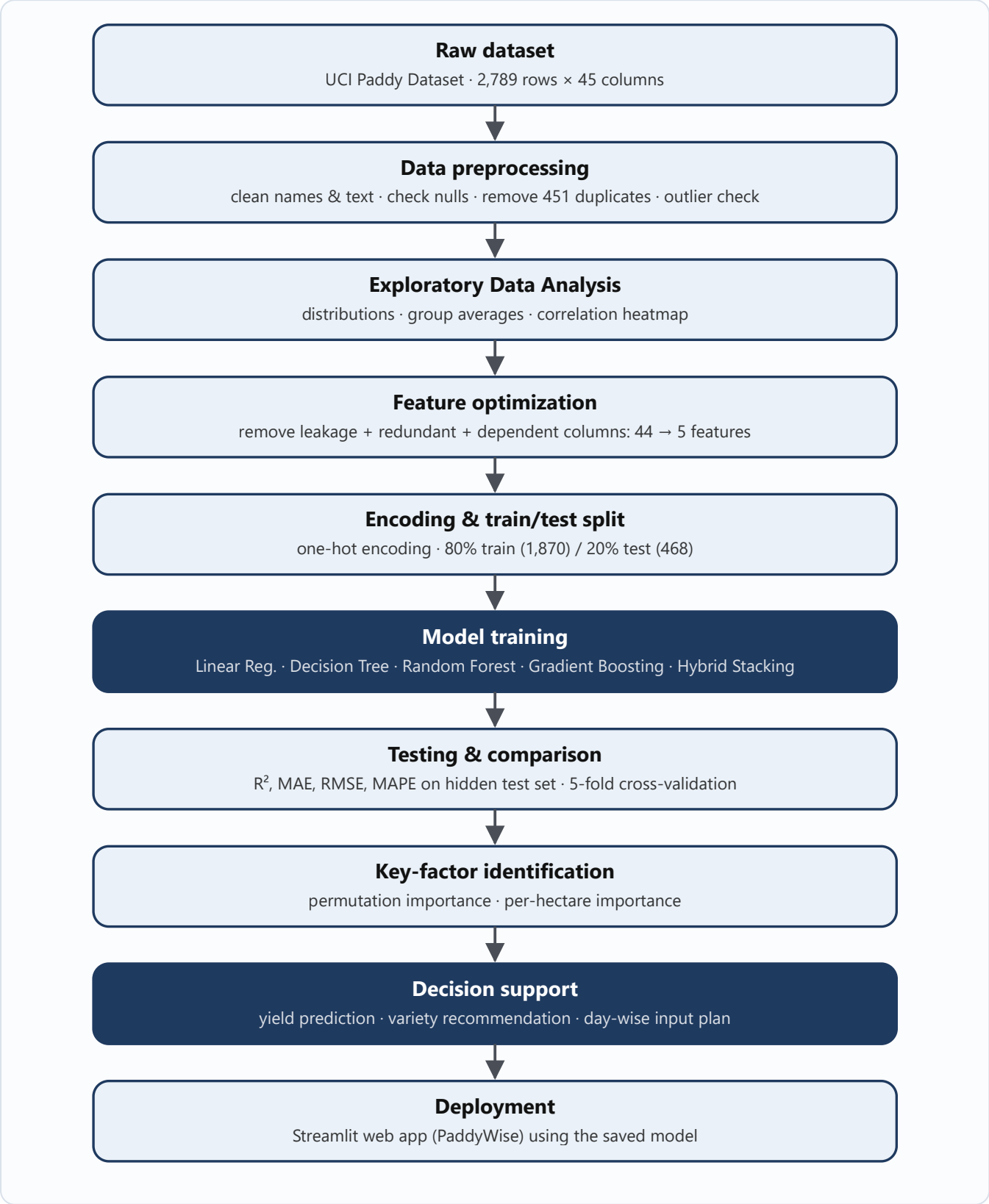


Figure 1. End-to-end methodology of the proposed framework.

3.1 Tools used

Stage	Library	Used for
Data handling	pandas, NumPy	loading, cleaning, duplicates, one-hot encoding, arithmetic
Visualisation	Matplotlib, Seaborn, Plotly	EDA plots, result charts, interactive app charts
Machine learning	scikit-learn	train_test_split, KFold, cross_val_score, 5 regressors, StackingRegressor, metrics, permutation importance
Persistence	joblib	saving the trained model (paddy_model.pkl)
Deployment	Streamlit	the PaddyWise web application

4. Data preprocessing

Raw data usually contains formatting problems, repeated records and errors. Preprocessing makes the data reliable before any model sees it. Figure 2 shows the steps.

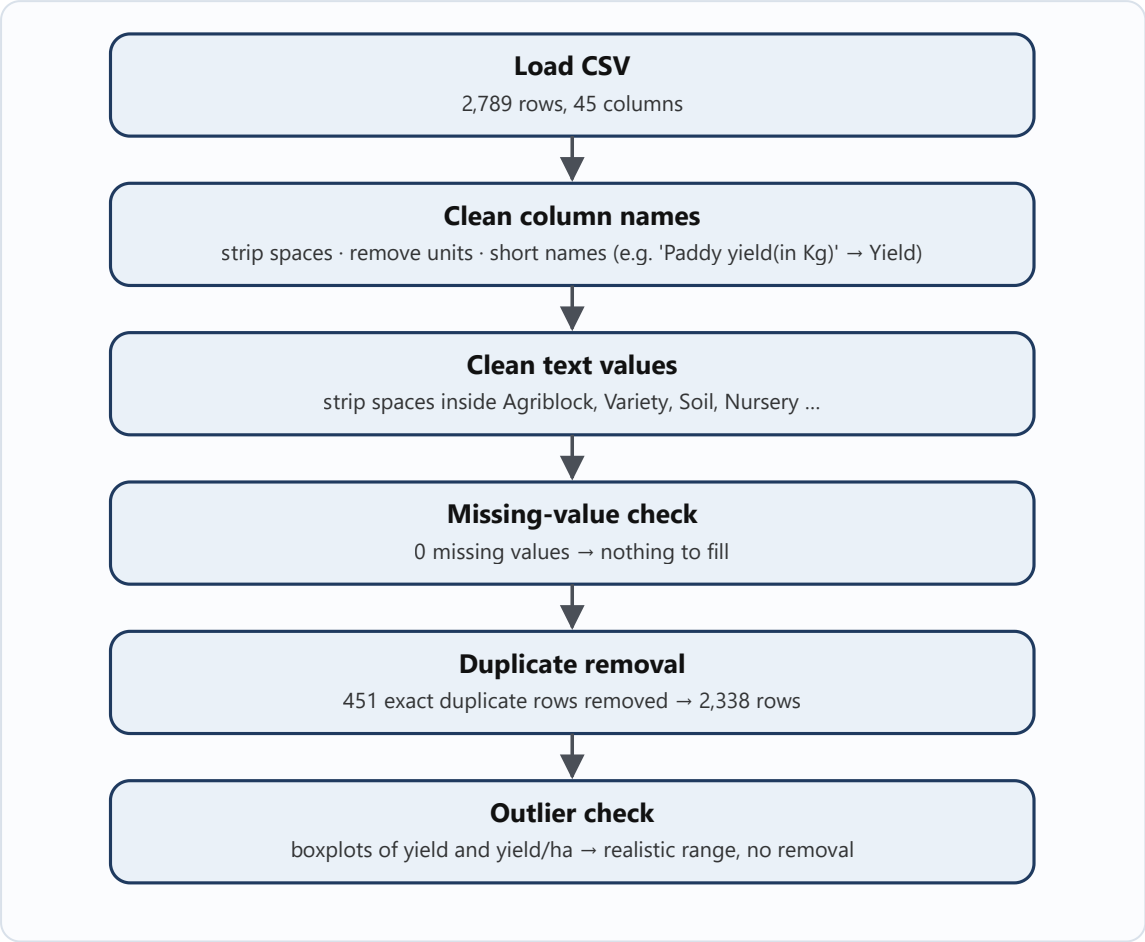


Figure 2. Preprocessing pipeline.

4.1 Cleaning column names and text

Several headers contained trailing spaces (e.g. `'Hectares '`) and long units (e.g. `'Paddy yield(in Kg)'`). We stripped the spaces, removed the bracketed units and renamed the columns to short names (`Yield`, `Soil`, `Seedrate` …). Text values such as block and variety names were stripped too, so that "Cuddalore " and "Cuddalore" are not treated as two different blocks.

4.2 Missing values

`df.isnull().sum()` returned **0** for every column, so no filling was needed.

4.3 Duplicate removal

We found **451 exact duplicate rows** (16.2% of the data). If a row appears in both the training and the test set, the model is tested on a row it has already seen, which inflates the score. Removing them gives **2,338 unique rows** and an honest evaluation.

4.4 Outliers

Boxplots of total yield and yield per hectare (Figure 3) show no extreme points. Yield per hectare lies between 5,410 and 6,469 kg/ha, a realistic range, so no rows were removed.

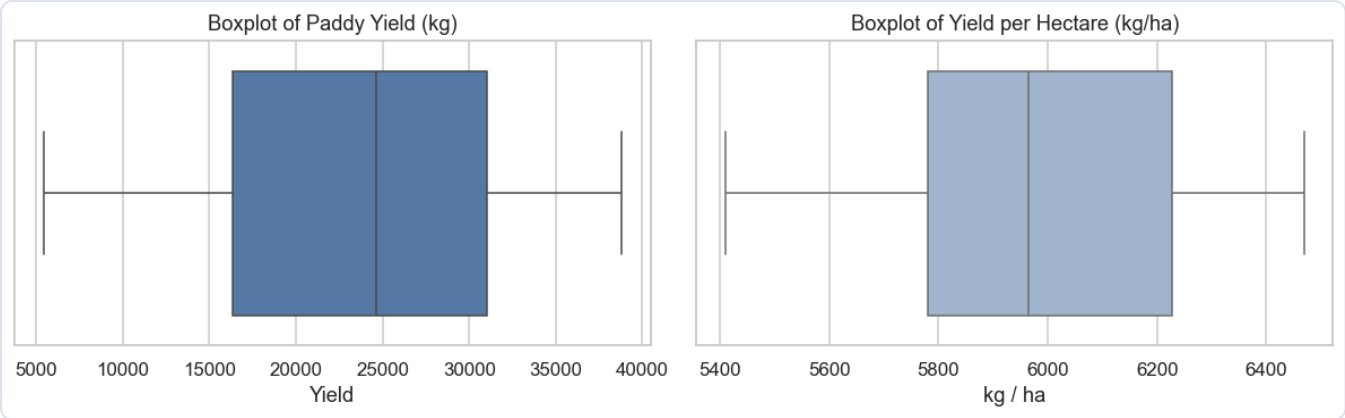


Figure 3. Outlier check: total yield (left) and yield per hectare (right).

4.5 Categorical values

Column	Categories (rows after cleaning)
Agriblock	Sankarapuram (502), Kurinjipadi (401), Cuddalore (364), Panruti (360), Chinnasalem (358), Kallakurichi (353)
Variety	ponmani (895), delux ponni (832), CO_43 (611)
Soil	clay (1268), alluvial (1070)
Nursery	dry (1255), wet (1083)

5. Exploratory data analysis (EDA)

EDA means studying the data with statistics and graphs before modelling, to understand which factors matter and to spot problems.

5.1 Distribution of yield

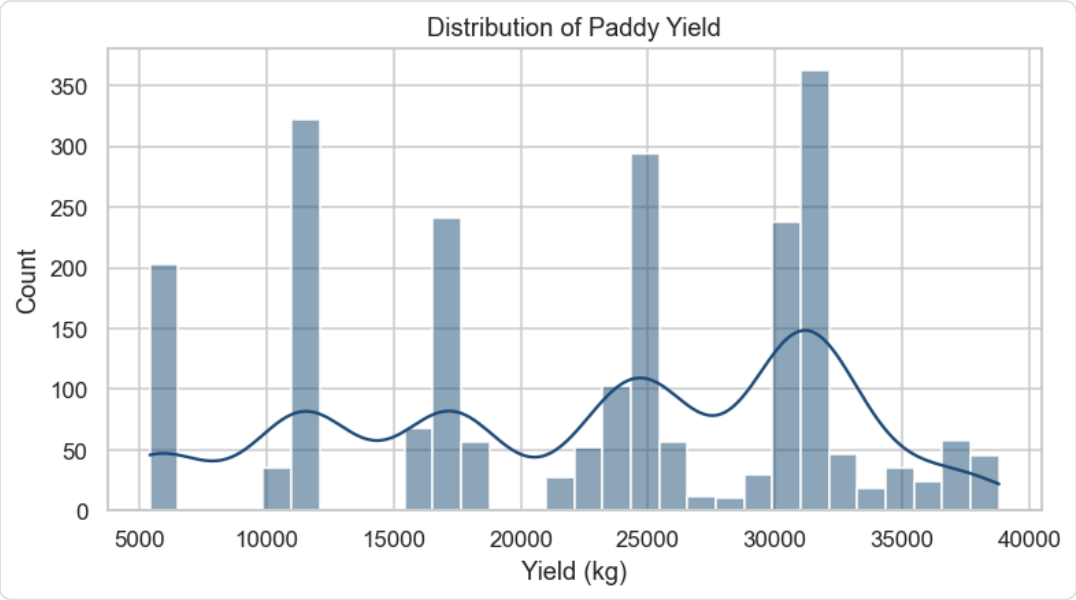


Figure 4. Distribution of total paddy yield. The peaks correspond to the six farm sizes (1–6 ha).

5.2 Yield versus farm area

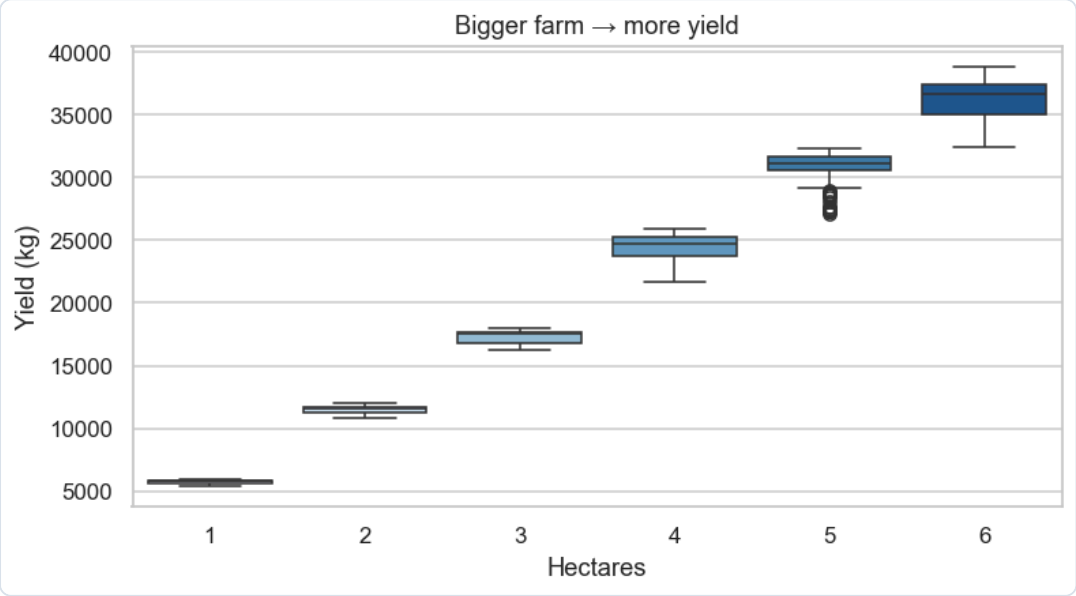


Figure 5. Yield grows almost linearly with farm area (correlation 0.994).

The correlation between Hectares and Yield is **0.994**, the strongest relationship in the data.

5.3 Yield per hectare by category

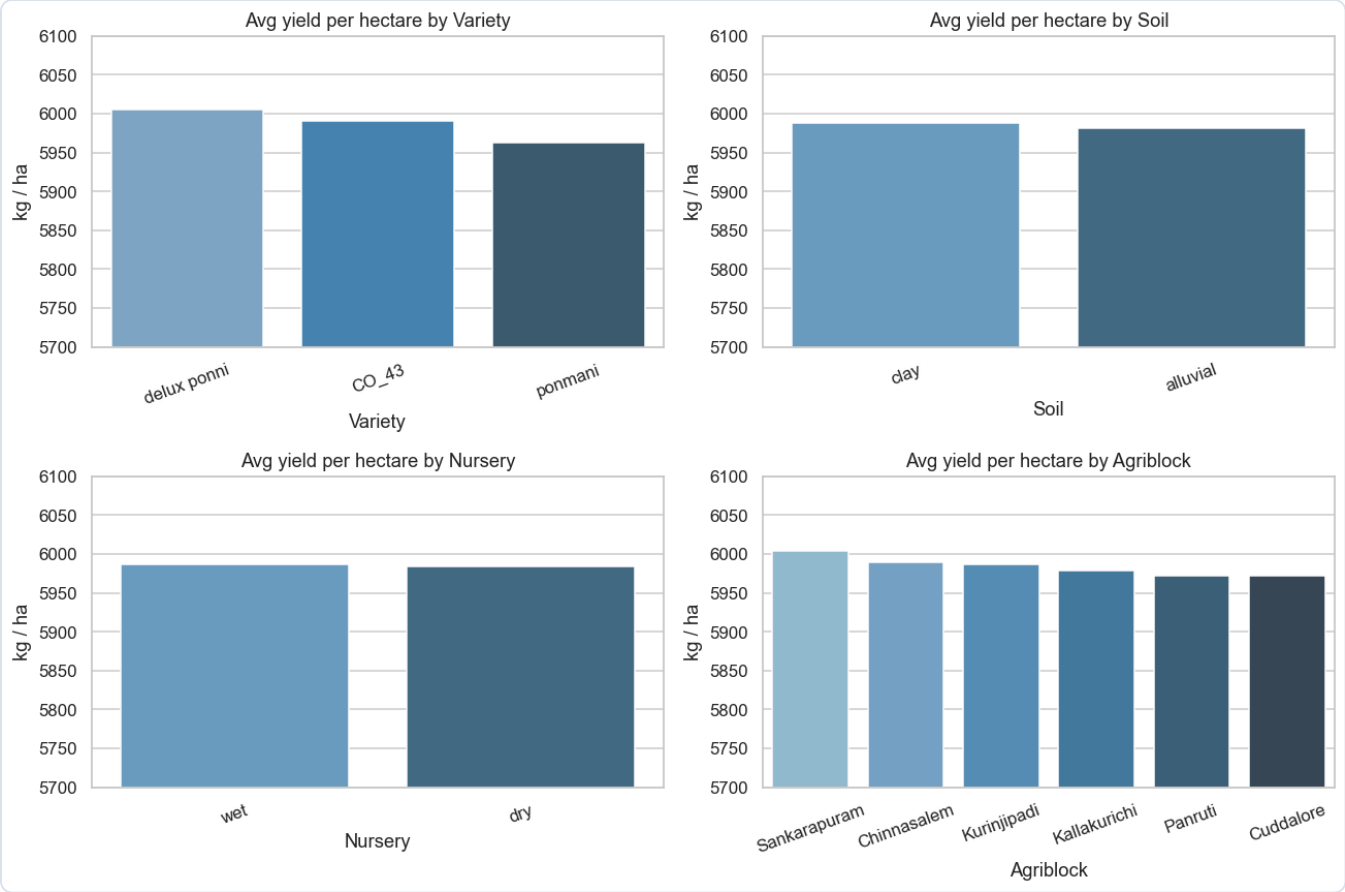


Figure 6. Average yield per hectare by variety, soil, nursery type and agri-block (y-axis starts at 5,700 to make small differences visible).

Differences between categories are small (tens of kg/ha) compared with the effect of area. Delux Ponni and CO 43 average slightly more than Ponmani, and Sankarapuram has the highest block average.

5.4 Agri-block climate profiles

Every farm in the same block shares exactly the same weather values, so each block has one climate profile:

Agri-block	Season rainfall (mm)	Avg max temp (°C)	Avg min temp (°C)	Avg humidity (%)	Avg yield (kg/ha)	Farms
Sankarapuram	430	32.2	18.1	82	6004	502
Chinnasalem	430	32.4	15.9	85	5990	358
Kurinjipadi	435	33.0	17.8	80	5987	401
Kallakurichi	429	32.2	17.0	87	5978	353
Panruti	429	32.9	18.4	86	5973	360
Cuddalore	435	32.0	16.5	81	5972	364

5.5 Correlation heatmap

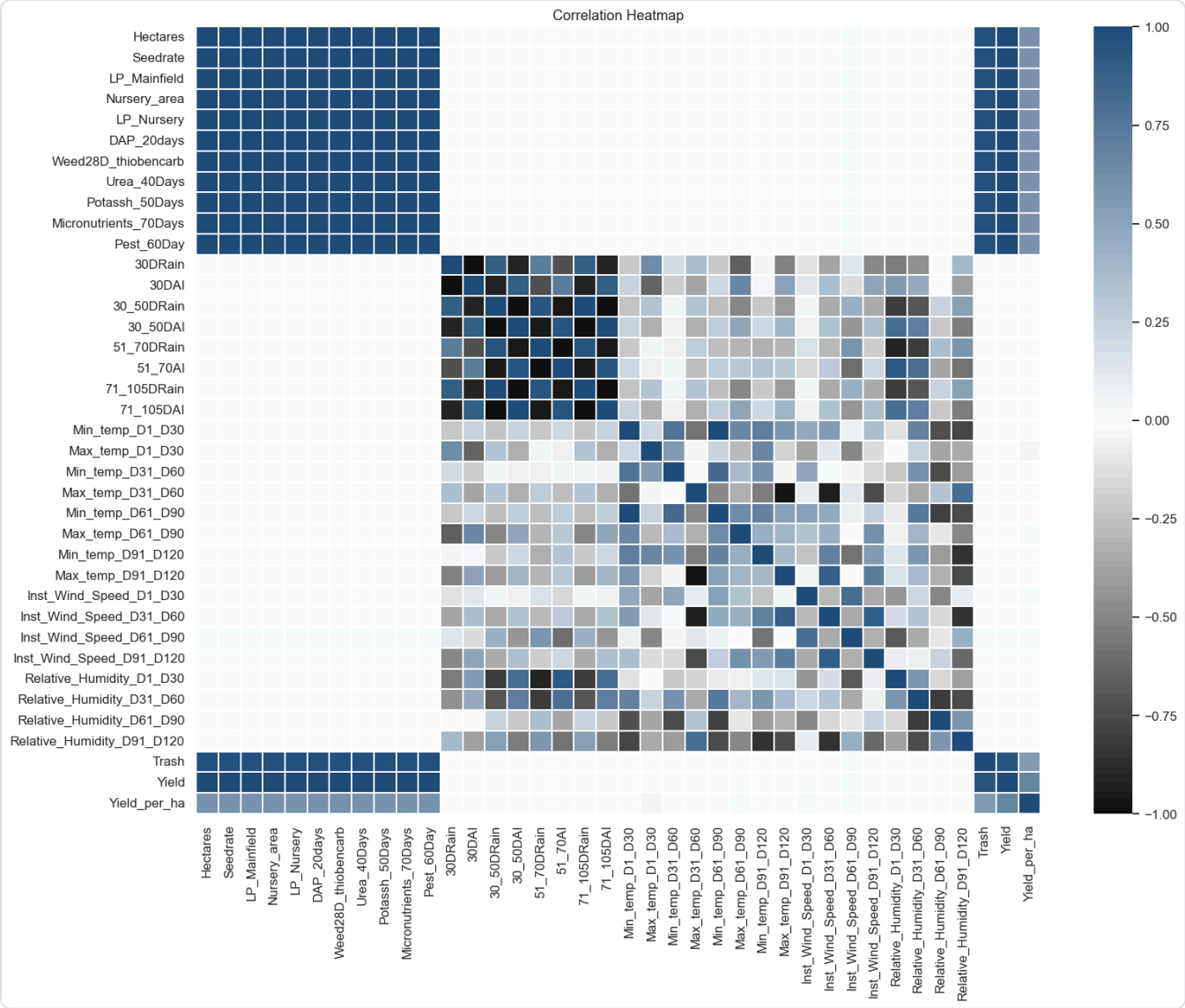


Figure 7. Correlation heatmap of numeric columns. The solid block shows that all input-dose columns are perfectly correlated with Hectares.

> **EDA findings.** (1) Farm area dominates total yield. (2) All fertilizer, seed and pesticide columns are Hectares × a fixed dose, so they repeat the same information. (3) Weather is constant within each agri-block. (4) Variety, soil and nursery have small but real effects on productivity.

6. Feature optimization

Feature optimization keeps only useful, non-repeated, legitimately available inputs. A smaller feature set is faster, easier to explain, less likely to overfit and simpler for a farmer to fill in. We applied three filters (Figure 8).

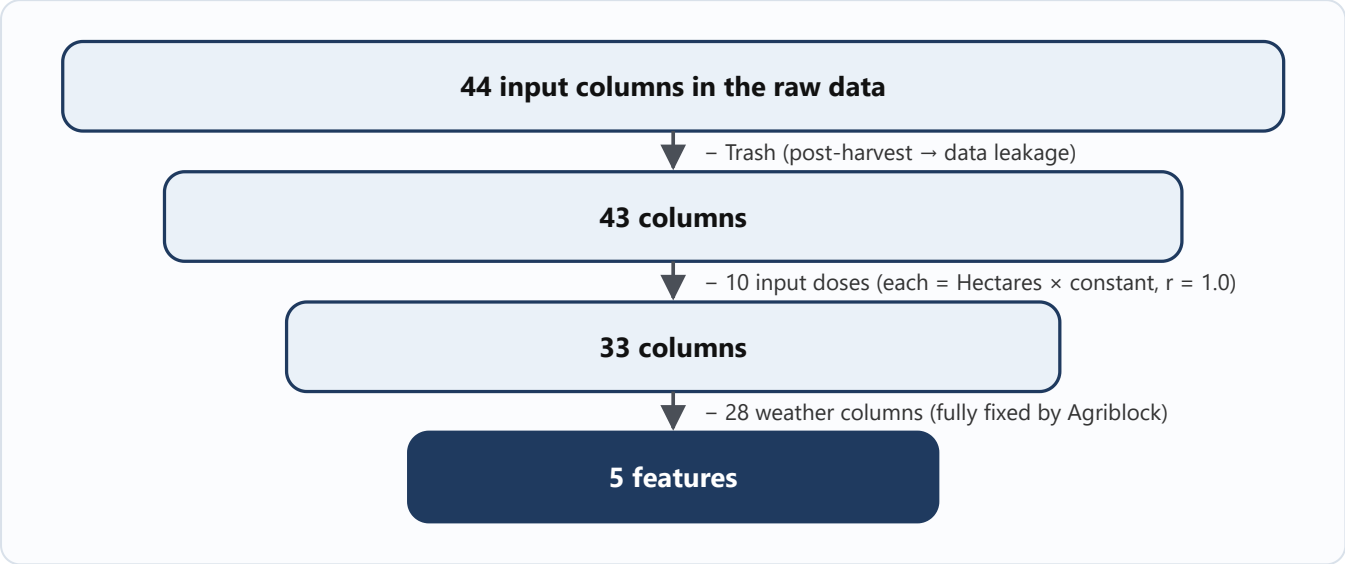


Figure 8. Feature-optimization funnel: 44 → 5 features.

6.1 Filter 1: data leakage

Trash (straw bundles) is measured *after* harvest. A farmer cannot know it when asking for a prediction, so using it would be "cheating" (target leakage). It was removed.

6.2 Filter 2: redundancy (correlation > 0.95)

Seed rate, nursery area, both land-preparation inputs, DAP, thiobencarb, urea, potash, micronutrients and pesticide all had correlation **1.000** with Hectares. Dividing each by Hectares gives a constant: the standard dose per hectare. They carry no information beyond Hectares, so they were removed, and their per-hectare doses were saved for the advice module (Chapter 11).

6.3 Filter 3: functional dependency on Agriblock

For all 28 weather columns, the number of distinct values *inside* any agri-block is exactly **1** (`df.groupby("Agriblock")[weather_cols].nunique().max() = 1`). Knowing the block therefore fixes all the weather values, so the single categorical feature **Agriblock** represents location and microclimate together.

6.4 Final features and encoding

> **Final features:** Hectares · Agriblock · Variety · Soil · Nursery

Text features were **one-hot encoded** (each category becomes a 0/1 column), giving 14 model columns: Hectares, 6 block columns, 3 variety columns, 2 soil columns and 2 nursery columns.

6.5 Did optimization cost accuracy?

Feature set	Model input columns	Test R^2 (hybrid model)
All original columns (except Trash)	70	0.9903
Optimized features	**14**	**0.9903**

Accuracy is identical with **80 % fewer columns**, which confirms that the removed columns were redundant.

7. Model development

7.1 Train/test split

The 2,338 rows were shuffled with a fixed seed (42) and split into **80 % training (1,870 rows)** and **20 % testing (468 rows)**. The test rows are hidden during training and used only for the final evaluation.

7.2 Models compared

Model	How it works	Settings
Linear Regression	Fits $\hat{y} = b + \Sigma\, w_i x_i$ by least squares (normal equation).	default
Decision Tree	Asks yes/no questions; each split minimises the squared error of the two groups; a leaf predicts its average.	max_depth = 8
Random Forest	Many trees on bootstrap samples; prediction = average of trees (lower variance).	300 trees
Gradient Boosting	Starts from the mean; each new small tree fits the remaining errors and is added with learning rate η.	300 trees, depth 3, η = 0.05
Hybrid Stacking	Three base models combined by a meta-model trained on their out-of-fold predictions.	see 7.3

7.3 The proposed hybrid stacking model

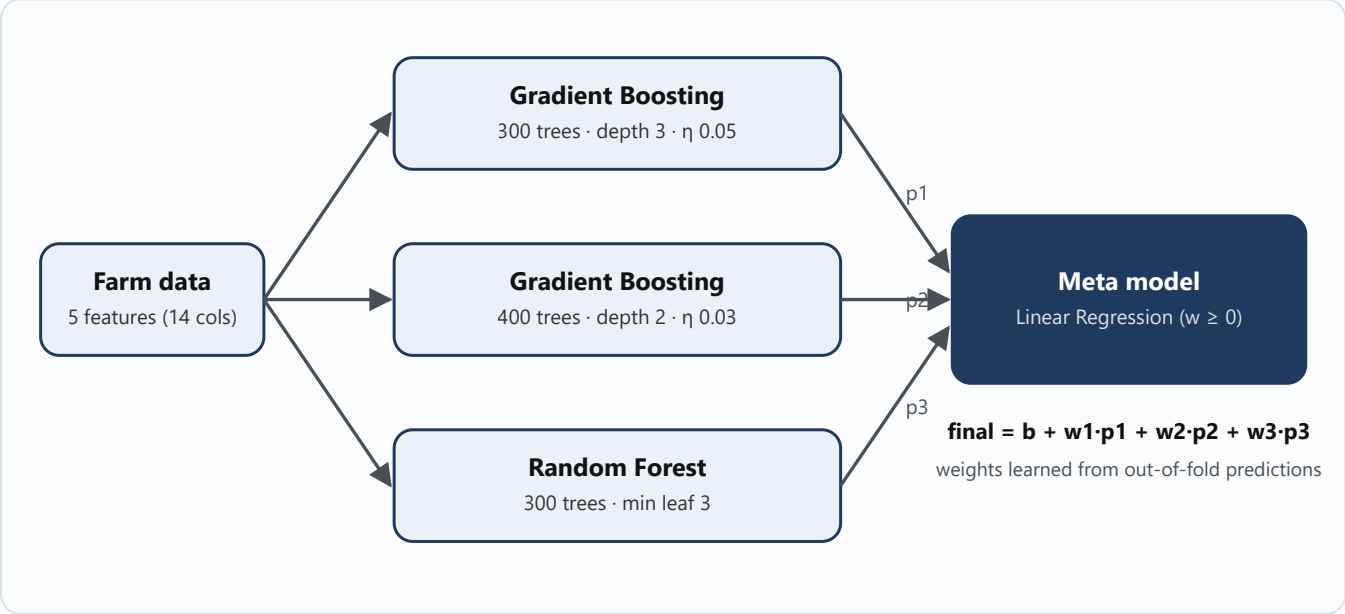


Figure 9. Architecture of the hybrid stacking regressor.

Base models. A deeper Gradient Boosting model (depth 3) captures interactions such as area × location; a shallower one (depth 2) captures smooth trends; a Random Forest gives stable, low-variance predictions.

Meta-model. A Linear Regression with **non-negative weights** learns final = $b + w_1p_1 + w_2p_2 + w_3p_3$. Negative weights would mean "do the opposite of a good model", which is not meaningful, so they are not allowed.

Training procedure.

1. Split the training data into 5 **shuffled** folds.
2. For each fold, train every base model on the other 4 folds and predict this fold. This gives **out-of-fold predictions**: every training farm gets a prediction from models that never saw it.
3. Train the meta-model on these out-of-fold predictions so that it learns realistic trust weights.
4. Retrain all base models on the full training set for final use.

> **Why shuffle the folds?** The CSV is sorted by farm size. Without shuffling, a fold could contain only 6-hectare farms and the base models trained without it would never see that size.

Learned weights. The meta-model learned w = [0.357, 0.643, 0.000] (GB depth 3, GB depth 2, Random Forest) with intercept -10.4. The Random Forest received weight 0: it was slightly less accurate and its errors overlapped with those of the boosting models, so it added no new information. Stacking selects useful models automatically.

8. Testing and model comparison

Testing answers the question *"how well will the model work on farms it has never seen?"* Figure 10 shows the procedure.

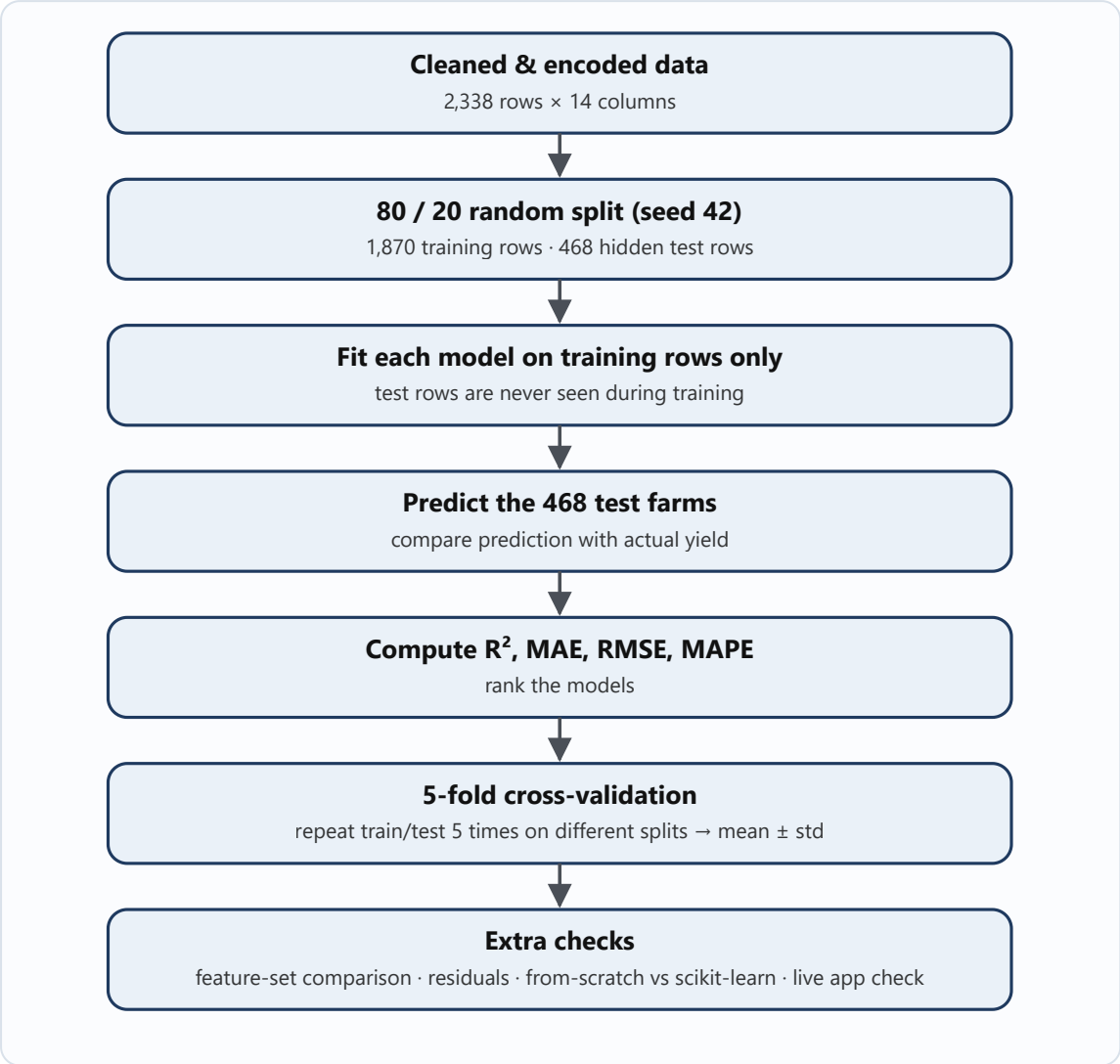


Figure 10. Testing and comparison procedure.

8.1 Evaluation metrics

Metric	Formula	Meaning	Good value
R^2	$1 - \Sigma(y - \hat{y})^2 / \Sigma(y - \bar{y})^2$	share of yield variation explained (vs always guessing the mean)	close to 1
MAE	$(1/n)\ \Sigma\ \lvert y - \hat{y} \rvert$	average error in kg	low
RMSE	$\sqrt{[(1/n)\ \Sigma\ (y - \hat{y})^2]}$	error that penalises big mistakes more	low
MAPE	$(100/n)\ \Sigma\ \lvert y - \hat{y} \rvert\ /\ y$	average error in %	low
Accuracy	$100 - \text{MAPE}$	easy-to-read "percentage correct"	high

8.2 Hold-out test results (468 unseen farms)

Model	R^2	MAE (kg)	RMSE (kg)	MAPE (%)	Accuracy (100–MAPE)
Hybrid Stacking	**0.9903**	**658**	**913**	**2.89**	**97.11%**
Gradient Boosting	0.9902	665	917	2.93	97.07%
Decision Tree	0.9893	687	957	3.03	96.97%
Random Forest	0.9891	694	966	3.07	96.93%
Linear Regression	0.9875	770	1,033	3.66	96.34%



Figure 11. Test-set comparison of the five models: R^2 (left) and MAE (right).

8.3 Cross-validation (stability check)

One split might be lucky. In 5-fold cross-validation the data is split five different ways; each model is trained and tested five times and the scores are averaged.

Model	CV mean R^2	CV std R^2	CV mean MAE (kg)
Hybrid Stacking	**0.9916**	0.0011	**611**
Gradient Boosting	0.9915	0.0011	617
Decision Tree	0.9904	0.0009	651
Random Forest	0.9901	0.0008	664
Linear Regression	0.9887	0.0015	744

The hybrid model's R^2 in the five folds was 0.9903, 0.9912, 0.9910, 0.9921 and 0.9935: consistently high, with a very small standard deviation.

8.4 Prediction quality and residuals

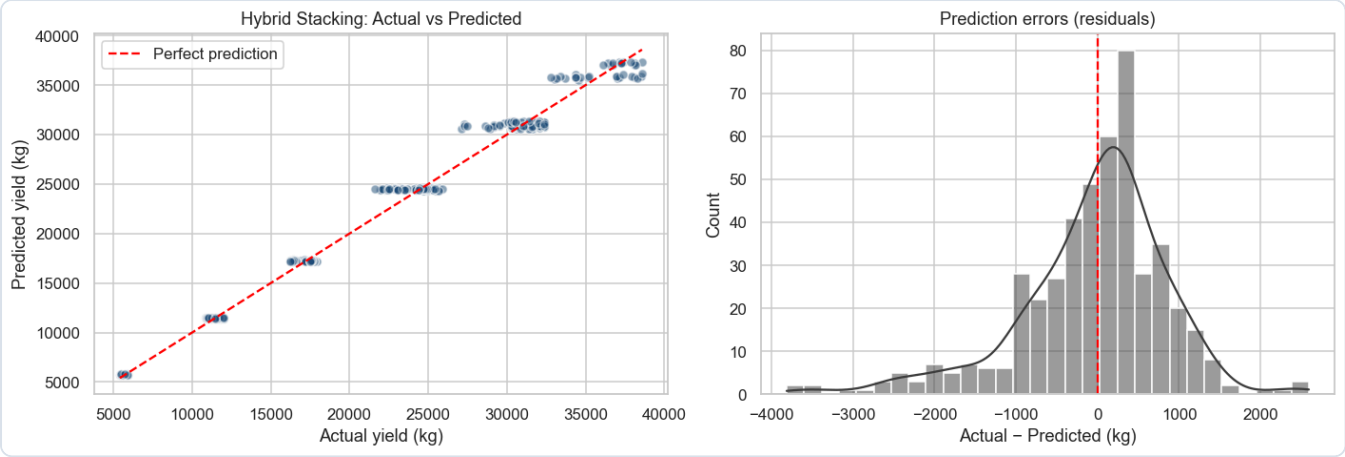


Figure 12. Hybrid model: actual vs predicted yield (left) and distribution of errors (right) on the test set.

Points lie on the diagonal and the residuals are centred on zero, so the model is accurate and unbiased. 80 % of test errors fall between **-5.3 %** and **+3.5 %** of the prediction; this band is shown in the app as the "likely range".

8.5 Comparison summary

- All tree-based models beat Linear Regression, because the effects of area and block are not perfectly linear.
- The **Hybrid Stacking** model is best on the test set (R^2 0.9903, MAE 658 kg) **and** in cross-validation (R^2 0.9916, MAE 611 kg).
- Gains over the best single model are small because all good models are near the limit of the data: farms with identical conditions still differ by natural variation that no model can predict. The hybrid is never worse than its best member and chooses the mix automatically.
- Feature optimization (70 → 14 columns) did not change accuracy (Section 6.5).
- Our from-scratch NumPy implementation reproduces the hybrid's accuracy (Chapter 12).
- The web app re-checks the model live on random test farms (Chapter 13).

9. Key factors influencing paddy yield (dominant features)

9.1 Method: permutation importance

For each feature we randomly shuffle its values in the test set and measure how much R^2 drops. If shuffling a feature destroys accuracy, the model depends on it. One-hot columns were summed back into their original feature.

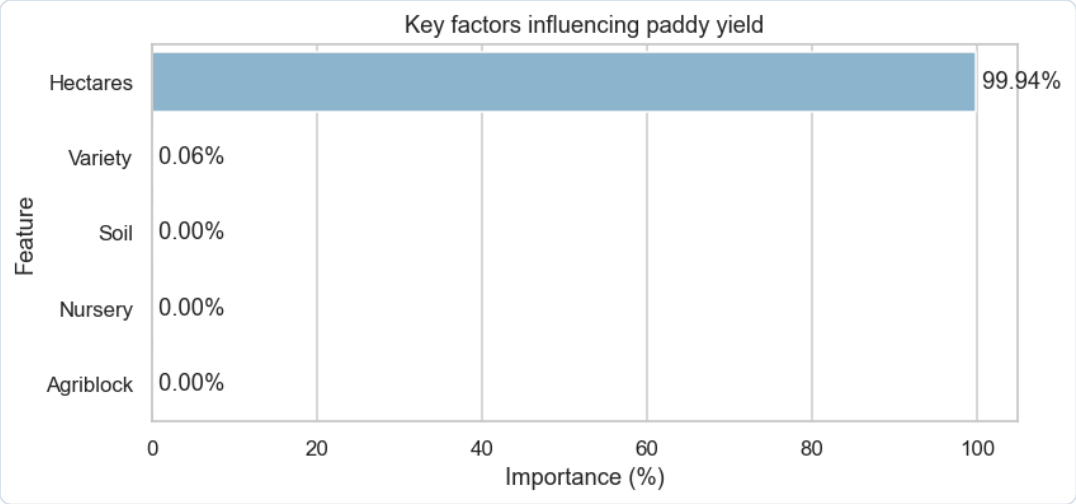


Figure 13. Permutation importance for total yield (hybrid model).

Farm area (Hectares) accounts for ≈ 99.9 % of the importance for total yield. This matches EDA: bigger farms produce proportionally more paddy, and all inputs scale with area.

9.2 What drives productivity (yield per hectare)?

To see the factors behind *productivity*, we trained a Random Forest on yield per hectare:

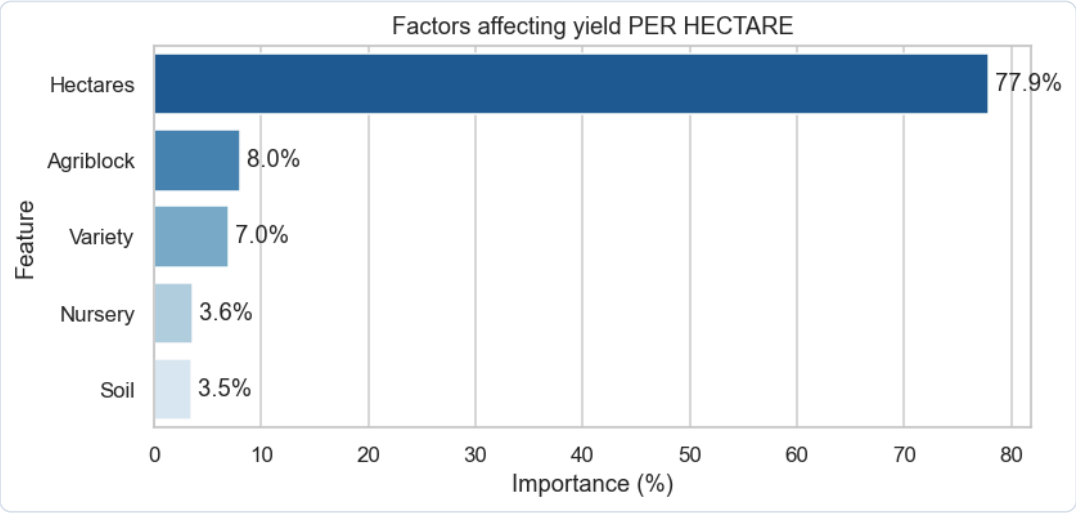


Figure 14. Feature importance for yield per hectare.

Rank	Factor	Importance	Interpretation
1	**Farm area**	77.9 %	productivity per hectare also changes with farm size (scale and management effect)
2	**Agri-block**	8.0 %	location and its microclimate (rainfall, temperature, humidity, wind)
3	**Variety**	7.0 %	genetic potential of CO 43 / Ponmani / Delux Ponni
4	**Nursery type**	3.6 %	wet vs dry nursery establishment
5	**Soil type**	3.5 %	clay vs alluvial

> **Dominant feature:** Farm area. **Most important controllable choice:** variety (with location fixed for a given farmer). Weather affects yield through the agri-block profile.

10. Decision support: variety recommendation

The decision-support layer turns predictions into a recommendation. Figure 15 shows the full logic. Only one step is machine learning (dark box); the others are transparent formulas.

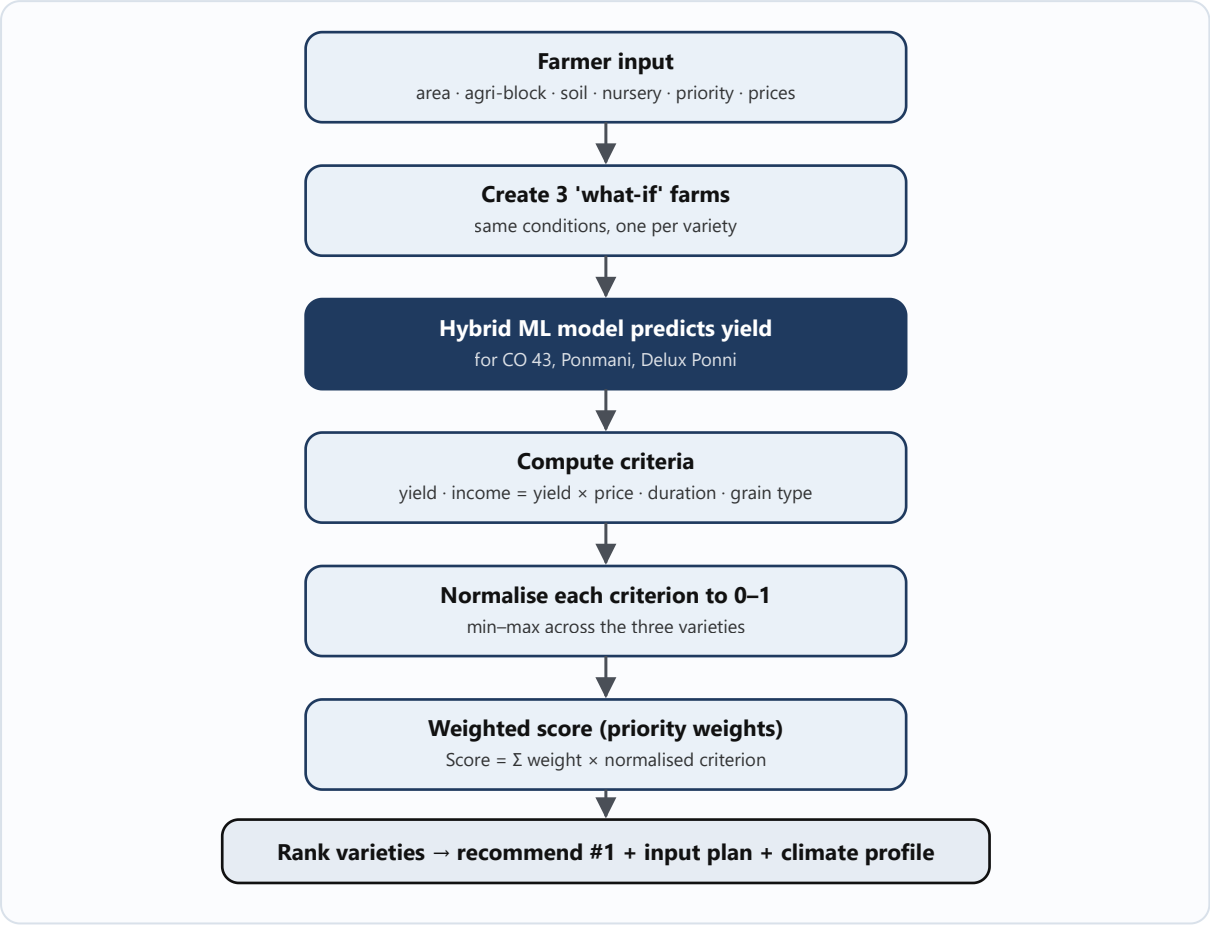


Figure 15. Recommendation engine.

10.1 Step-by-step

1. **Input:** farm area, agri-block, soil, nursery type, the farmer's priority and (optionally) market prices.
2. **What-if farms:** three copies of the farm are created, identical except for the variety.
3. **ML prediction:** the hybrid model predicts the yield of each copy, giving the expected yield of each variety under exactly the farmer's conditions.
4. **Criteria:** income = predicted yield × price; duration and grain type come from the variety profile.
5. **Normalisation:** each criterion is rescaled to 0–1 across the three varieties:

   $\text{norm}(x) = (x - \min) \,/\, (\max - \min)$. For duration, shorter is better, so we use −duration.

6. **Weighted score:** $\text{Score} = w_y \cdot \text{Yield} + w_i \cdot \text{Income} + w_d \cdot \text{Shortness} + w_g \cdot \text{FineGrain}$
7. **Ranking:** the variety with the highest score is recommended; the others are shown as alternatives.

10.2 Priority weights

Farmer priority	Yield	Income	Shorter duration	Fine grain
Maximum yield	70%	20%	10%	0%
Maximum profit	20%	70%	10%	0%
Short duration	30%	20%	50%	0%
Fine grain	20%	30%	0%	50%

> The weights express the farmer's priority; they are design choices, not learned from data, because the dataset contains no record of which variety a farmer "should" have chosen. This weighted-sum approach is a standard multi-criteria decision-making (MCDM) method. A farmer or extension officer can change the weights, and future work could derive them with AHP or from real farmer feedback.

10.3 Worked example

Farm: **4 ha, Cuddalore, clay soil, wet nursery**, priority = **Maximum profit** (weights: yield 20 %, income 70 %, duration 10 %).

Rank	Variety	Predicted yield (kg)	Income (₹)	Duration	Yield pts	Income pts	Duration pts	Score /100
1	Delux Ponni	24,412	659,113	140	14.4	70.0	7.5	92
2	CO 43	24,439	537,654	135	20.0	12.5	10.0	43
3	Ponmani	24,341	511,169	155	0.0	0.0	0.0	0

CO 43 has the highest predicted yield, but Delux Ponni's fine grain earns a higher price (₹27 vs ₹22/kg), so its income is far higher. With profit as the priority, **Delux Ponni** wins. Expected yield 24,412 kg (likely range 23,120–25,264 kg), estimated income ₹659,113.

Same farm, different priorities

Farmer priority	Recommended	Runner-up	Winning score
Maximum yield	**CO 43**	Delux Ponni	84/100
Maximum profit	**Delux Ponni**	CO 43	92/100
Short duration	**CO 43**	Delux Ponni	84/100
Fine grain	**Delux Ponni**	CO 43	94/100

The recommendation changes with what the farmer values. This is the "farmers' preferences" part of the problem statement.

11. Advice generation and day-wise recommendation plan

Besides the variety, the system gives practical advice: **how much of each input to apply and when**, plus the expected yield, the likely range, the income and the climate profile of the block.

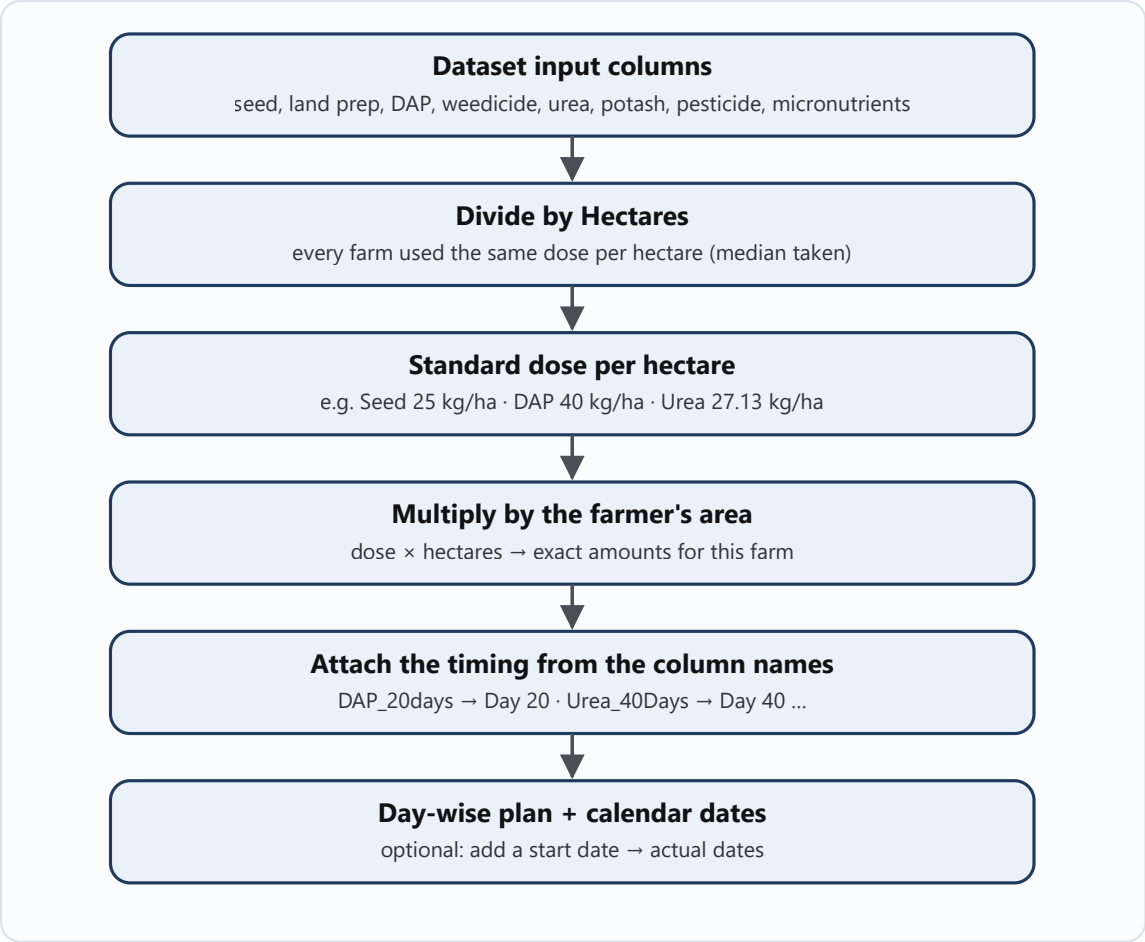


Figure 16. How the input plan is generated from the data.

11.1 Where the doses come from

In Chapter 6 we found that every input column equals Hectares × a fixed amount. Dividing by Hectares therefore recovers the **standard management practice per hectare** used on these farms. Multiplying it by the farmer's area gives the amounts for that farm. The timing (day 20, 28, 40 …) is part of the column names in the dataset.

Input (dataset column)	Dose per hectare	For 4 ha	When
Seedrate	25 kg	100.0 kg	Day 0
Nursery_area	20 cents	80.0 cents	Day 0
LP_Nursery	1 t	4.0 t	Day 0
LP_Mainfield	12.5 t	50.0 t	Before transplanting
DAP_20days	40 kg	160.0 kg	Day 20
Weed28D_thiobencarb	2 units	8.0 units	Day 28
Urea_40Days	27.13 kg	108.5 kg	Day 40
Potassh_50Days	10.38 kg	41.5 kg	Day 50
Pest_60Day	600 ml	2,400.0 ml	Day 60
Micronutrients_70Days	15 kg	60.0 kg	Day 70

11.2 Day-wise timeline

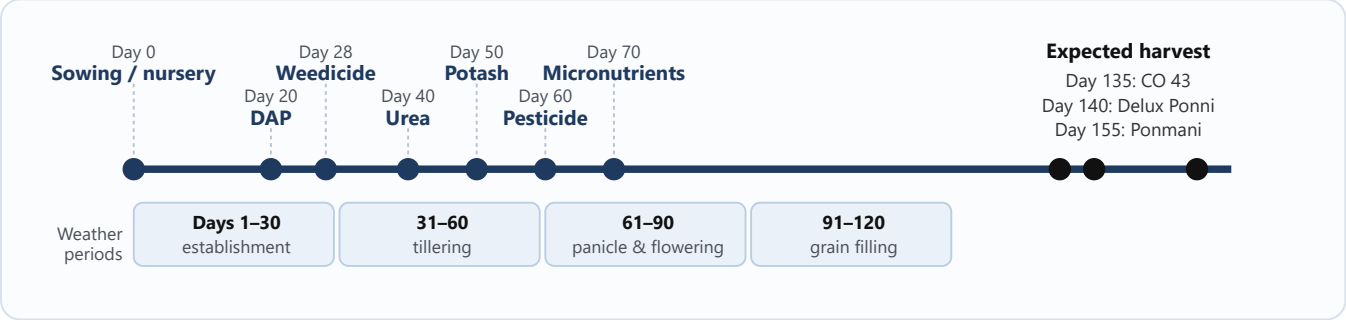


Figure 17. Day-wise crop calendar with the weather periods recorded in the dataset and the expected harvest of each variety.

The four weather periods in the dataset (days 1–30, 31–60, 61–90, 91–120) line up with the crop stages: nursery and establishment, tillering, panicle initiation and flowering, and grain filling. The input schedule follows these stages: phosphorus early for roots (DAP, day 20), weed control before the canopy closes (day 28), nitrogen for tillering (urea, day 40), potassium for grain filling and stem strength (day 50), pest protection at peak vegetative growth (day 60), and micronutrients before flowering (day 70).

11.3 Date-wise calendar (example)

If the farmer gives a start date, each "Day N" becomes a calendar date. Example for a 4-hectare farm sown on **1 August 2026**:

Day	Date (if sown 1 Aug 2026)	Activity	Amount for 4 ha
0	01 Aug 2026	Seed for nursery	100.0 kg
0	01 Aug 2026	Nursery area to prepare	80.0 cents
0	01 Aug 2026	Land preparation – nursery	4.0 t
20	21 Aug 2026	DAP fertilizer (basal phosphorus)	160.0 kg
28	29 Aug 2026	Thiobencarb weedicide	8.0 units
40	10 Sep 2026	Urea top-dressing (nitrogen)	108.5 kg
50	20 Sep 2026	Muriate of potash (potassium)	41.5 kg
60	30 Sep 2026	Pesticide spray	2,400.0 ml
70	10 Oct 2026	Micronutrient mixture	60.0 kg
135	14 Dec 2026	Expected harvest – CO 43	—
140	19 Dec 2026	Expected harvest – Delux Ponni	—
155	03 Jan 2027	Expected harvest – Ponmani	—

"Day" is counted from the start of the crop season as recorded in the dataset. Units for the weedicide are reported exactly as in the dataset. Doses reflect the practice recorded on the farms in the dataset; local agricultural department guidance should take priority where it differs.

11.4 Summary of the advice given to a farmer

1. **Which variety to grow**, ranked for the chosen priority, with the reason (score breakdown).
2. **Expected yield** (kg and tonnes), yield per hectare, and a likely range.
3. **Estimated income** at the entered market prices.
4. **Crop duration and expected harvest** for each variety.
5. **Day-wise input plan**: seed, nursery area, land preparation, DAP, weedicide, urea, potash, pesticide and micronutrients, in exact amounts for the farm.
6. **Season climate** of the block (rainfall, temperature, humidity) for planning.

12. From-scratch implementation and verification

To show a full understanding of the algorithms, the complete hybrid model was re-implemented using **only NumPy** (notebook `Hybrid_Model_From_Scratch.ipynb`): train/test split, metrics, Linear Regression (normal equation with pseudo-inverse), non-negative Linear Regression (coordinate descent), Decision Tree (exhaustive SSE split search with running sums), Random Forest (bootstrap + averaging), Gradient Boosting (sequential residual fitting), K-Fold and Stacking.

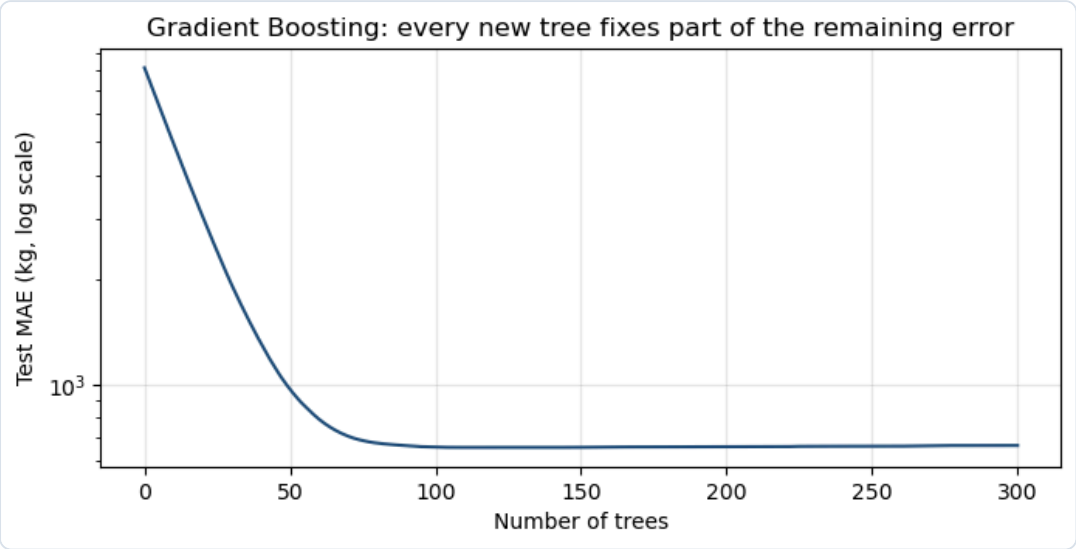


Figure 18. Our from-scratch Gradient Boosting: the test error falls as trees are added (log scale).

12.1 Results on the same 468 test farms

Model	R^2	MAE (kg)	MAPE (%)
scikit-learn Hybrid (final model)	0.9903	658	2.89
Our from-scratch Hybrid (NumPy)	**0.9902**	**661**	**2.91**
Ours · Gradient Boosting (depth 2)	0.9903	657	2.88
Ours · Gradient Boosting (depth 3)	0.9902	665	2.93
Ours · Random Forest	0.9894	686	3.02
Ours · Linear Regression	0.9875	770	3.66

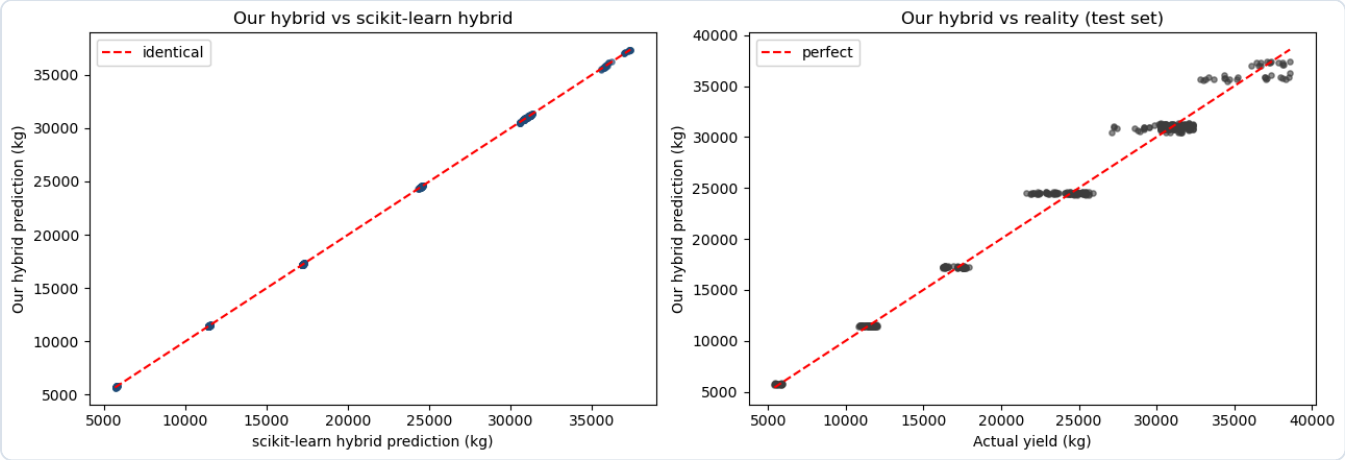


Figure 19. Left: our hybrid vs scikit-learn's hybrid. Right: our hybrid vs actual yield.

Our implementation reproduces the test split exactly and its predictions differ from scikit-learn's by only **0.08 % on average** (max 0.44 %). Both meta-models gave the Random Forest weight 0. The small differences come from random sampling, 100 vs 300 forest trees, and scikit-learn's slightly different split criterion (Friedman MSE).

13. Web application (PaddyWise)

The framework is deployed as an interactive Streamlit application (`app.py`) that loads the saved hybrid model.

Tab	What it does
Yield Predictor	Live prediction as inputs change: yield, yield/ha, likely range, number of real farms with identical conditions and their actual average; histogram placing the prediction among real farms; **live check on 6 random unseen test farms** (actual vs predicted vs error %).
Variety Advisor	Recommended variety for the chosen priority, ranked cards for all varieties, score-breakdown chart, yield-vs-farm-size chart, explanation of how the recommendation is made with the current weights, climate profile, day-wise input plan.
Data Insights	Interactive EDA charts, key factors, feature-optimization summary.
Model Performance	Live test-set metrics, actual-vs-predicted plot, model comparison, stacking architecture.
About	Problem statement, pipeline, team, tech stack, dataset citation.

Running: `pip install -r requirements.txt` then `streamlit run app.py`. The app can be deployed free on Streamlit Community Cloud from a GitHub repository.

14. Discussion and limitations

- **Data structure.** Inputs are exact multiples of area and weather is fixed per block, so the effective information is five features. This limits how much management practices can be studied separately from farm size.
- **Whole-hectare farms only.** The data covers 1–6 ha in whole numbers; predictions outside this range are extrapolations, so the app restricts the inputs.
- **Small variety differences.** Predicted yields of the three varieties differ by tens of kg/ha, so the recommendation is driven mainly by price, duration and grain type through the priority weights.
- **Hand-chosen weights and external variety data.** Priority weights and variety profiles (duration, grain, price) are not learned from the dataset; they are transparent and editable.
- **Irreducible noise.** Identical farms show different yields, so $R^2 \approx 0.99$ is close to the practical maximum.

15. Conclusion and future scope

We built a complete hybrid, feature-optimized, decision-support framework for paddy. Preprocessing removed 451 duplicates; feature optimization reduced 44 inputs to 5 without loss of accuracy; the hybrid stacking model achieved **R^2 = 0.9903, MAE = 658 kg, MAPE = 2.89 %** on unseen farms, with stable cross-validation and an independent from-scratch verification. Farm area is the dominant factor, followed by agri-block (microclimate), variety, nursery and soil. The decision-support layer recommends the best variety for each farmer's priority and produces a day-wise, area-scaled input plan, all available in an interactive web app.

Future scope: live weather and market-price APIs; more districts, seasons and varieties; learning priority weights from farmer feedback or AHP; soil-test (NPK) based fertilizer advice; a regional-language mobile interface.

16. References

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Appendix A: Frequently asked viva questions

Question	Answer
Is this classification or regression?	Regression: the target (yield in kg) is a continuous number. The recommendation is a scoring rule on top of regression predictions.
Why remove duplicates?	A duplicate in both train and test lets the model be tested on data it has seen, inflating scores.
Why drop Trash?	It is measured after harvest: target leakage.
Why drop fertilizer and weather columns?	Fertilizer columns are Hectares × constant (r = 1.0); weather is fully determined by Agriblock. They add no new information; accuracy is unchanged.
What is one-hot encoding?	Turning a text category into 0/1 columns, one per category, so models can use it.
What is stacking?	An ensemble where base models' out-of-fold predictions train a meta-model that learns the best way to combine them.
What is the meta-model?	A non-negative Linear Regression whose inputs are the base models' predictions; its weights say how much to trust each model.
Why is the Random Forest weight 0?	It was slightly less accurate and its errors overlapped with the boosting models', so it added no new information; scikit-learn and our NumPy version both chose 0.
Why out-of-fold predictions?	Predictions on seen rows look too good; out-of-fold ones behave like predictions on new farms, so the learned weights are realistic.
What does $R^2 = 0.99$ mean?	The model explains 99 % of the variation in yield compared with always predicting the average.
Why not $R^2 = 1$?	Farms with identical conditions have different yields (natural variation that is not recorded).
Why cross-validation?	To check the score is not due to one lucky split; 5 different splits give a mean and standard deviation.
What is the dominant feature?	Farm area; for yield per hectare: area, then agri-block, variety, nursery, soil.
How is the variety recommended?	The ML model predicts each variety's yield for the farm; income = yield × price; criteria are normalised to 0–1 and combined with priority weights; the highest score wins.
Are the weights learned?	No. They encode the farmer's priority (a standard MCDM approach) and are editable; the dataset has no "correct variety" label to learn from.
Where does the day-wise plan come from?	Per-hectare doses recovered from the dataset (input ÷ hectares), multiplied by the farm area; timings come from the column names (day 20, 28, 40, 50, 60, 70).
How do you know your code is right?	Hidden test set + 5-fold CV + a from-scratch NumPy implementation that matches scikit-learn within 0.08 % + live checks on unseen farms in the app.